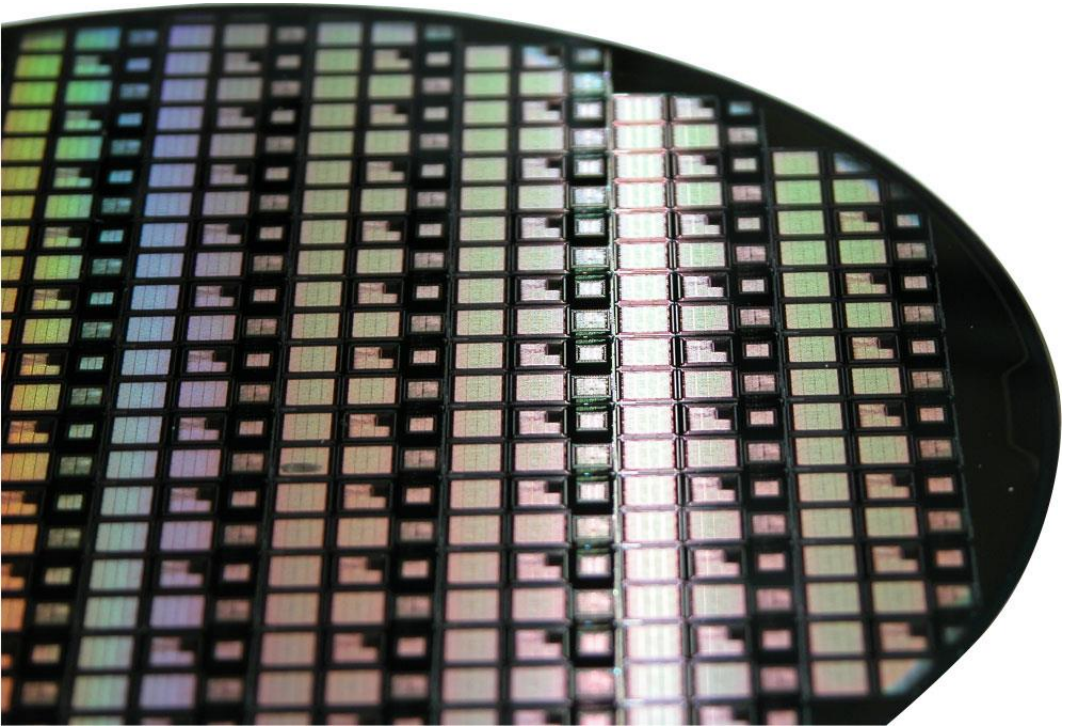
ETH zürich

Placement and routing



15 April 2025

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Where are we

Week	Short	Content
1	INTRO	Why this course, design flow, case for an open-source EDA flow
2	RTL	Refresher on System Verilog
3	Netlist	Logic synthesis, basics of digital circuit timing, Croc our example SoC we will use in class
4	ICT	Short summary of IC manufacturing
5	LEF	Standard cells, routing layers
6	DEF	Floorplanning, I/O rings, ESD structures, packaging
7	LIB	Timing, Clock trees, Process corners
8	SDF	Parasitics, extraction, issues in timing, crosstalk
9	**GDS**	**Placement, routing, chip finishing, DRC and LVS**
10	VCD	Power analysis and IR Drop
11	Project	Discussion on your final project, Q&A session
12	WGL	IC Testing, ATPG, JTAG, BIST
13	PPA	Properly reporting performance, power, area in IC Design
14	Final	Final thoughts, reserve

Previously we covered

- **Digital standard cells and how they are designed to be placed**
  - Basic logic functions realized in target technology
  - Physical layout finalized and timing characterized
  - Rectangular blocks with the height fixed, designed to be put side-by-side
- **Floorplan determines the area of our IC**
  - Reserve place for I/Os, macro cells and standard cells
  - The need for I/O pads, ESD protection, packaging
- **Introduction to timing**
  - Clock distribution, clock disciplines, process corners
- **Parasitics and all other issues that add to our problems**

Goals for this lecture

- **Placement of cells**
  - Introduction to algorithms and approaches used for placement
- **Routing**
  - How to connect all the cells together
  - Global and detail routing
- **Chip finishing**
  - Last steps before submitting the design
  - Filler, antenna cells
  - DRC, LVS
- **The GDS File**

Reading/references for this lecture

- **The slides in this lecture are taken from *https://www.eng.biu.ac.il/temanad/digital-vlsi-design/***

Prof. Adam Teman
Faculty of Engineering

Prof. Adam Teman | Teaching | Patents | Publications | Research Interests | Research Students | Research Tapeouts | Honors and Awards | Contact Information

Digital VLSI Design

Hebrew Name: מעגלי ומערכות וי.אל.אס.איי. דיגיטליים

BIU Course Number: 83-612

Complete Playlist on YouTube: English, עברית

intel

Preparation of these recorded lectures was kindly supported by Intel.

Find the original slides used in the lecture

- **Placement**
  http://www.eng.biu.ac.il/temanad/files/2018/12/Lecture-7-Placement.pdf

- **Routing**
  http://www.eng.biu.ac.il/temanad/files/2019/01/Lecture-9-Routing.pdf

- **Chip finishing**
  http://www.eng.biu.ac.il/temanad/files/2021/01/Lecture-11-Signoff-2020-21.pdf

GDS

Graphic Design System

GDS2 or GDSII is the standard file for physical layouts

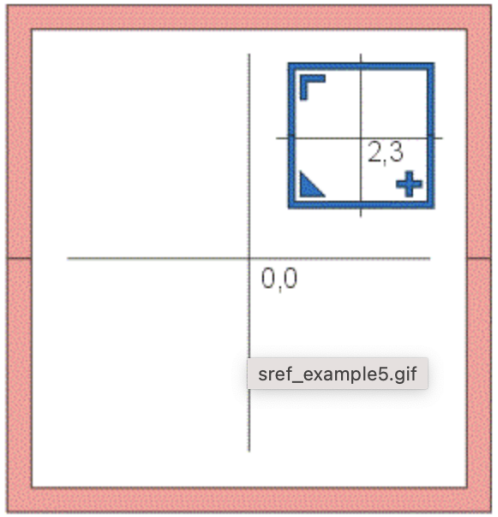
- **GDS2 has been in use for several decades**
  - It is an older binary format (unlike all others we have seen so far)
    - This was typical in the 80s and early 90s to save disk space
    - However, the format is not compressed (back in the day computing power was also an issue)
  - There is no official standard or body that maintains it
- **It is a format that contains ONLY geometric descriptions**
  - Rectangles
  - Polygons
  - Paths
- **For a description see**
  - `https://www.artwork.com/gdsii/gdsii/`

GDS layer and purpose pairs

- **GDS *layer* is used to describe different masks for the technology**
  - 8-bit number, no standard assignment
- **A second *purpose* can be used to differentiate different uses of layer**
  - For example: the actual drawing and the pin location on a metal could share the same layer but have different purposes
  - Also 8-bit number
- **In practice:**
  - Layer-purpose pairs give GDS 65k possible different masks
  - Every technology provider assigns these as part of the PDK
  - Tools convert the CAD database to the specific layer in the PDK

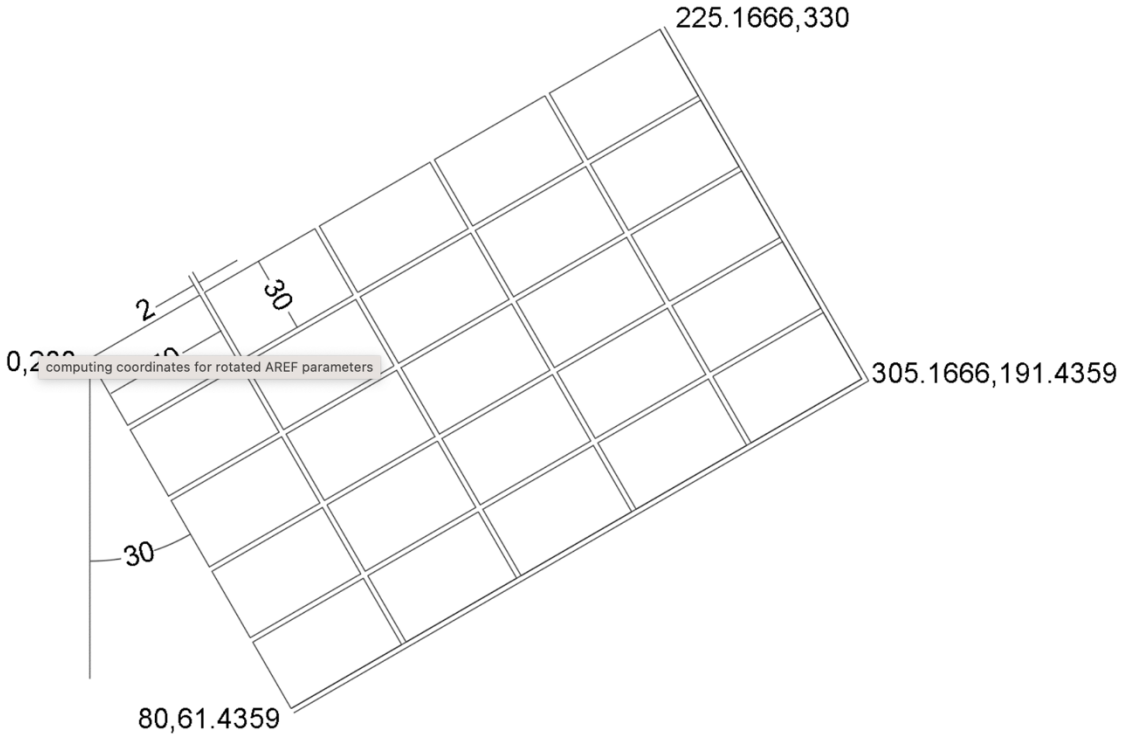
GDS supports hierarchy

- **Important aspect of GDS file**
- **Arrays, rotations and mirroring are also supported**




```
BGNSTR TOP
BOUNDARY
SREF CELL_A                    180 2  3
ENDSTR

BGNSTR CELL_A
BOUNDARY
BOUNDARY
BOUNDARY
BOUNDARY
ENDSTR
```



https://www.artwork.com/gdsii/gdsii

Next week

- **No lecture**
  - Easter holiday
- **Once we are back**
  - We talk about power